WORKED SOLUTION

STEP I 1993, Question 3

Pure • Diophantine Equations

> **STEP I 1993, Question 3**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/93-s1-q3

The key idea

Because $p \le q \le r$, the reciprocal $\frac{1}{p}$ is the *largest* of the three terms. So the sum is at most $\frac{3}{p}$, and that pins p down to a couple of values. Once p is fixed, the same trick on the two remaining terms bounds q, and then r is forced. The idea is to use the ordering to turn an equation (or an inequality) in three unknowns into a short finite check.

Solution

Part (i)

Since $1 \le p \le q \le r$, we have $\frac{1}{r} \le \frac{1}{q} \le \frac{1}{p}$. So

$$1 = \frac{1}{p} + \frac{1}{q} + \frac{1}{r} \le \frac{3}{p},$$

which gives $p \le 3$. Also $p \neq 1$, because then $\frac{1}{q} + \frac{1}{r} = 0$, which is impossible for positive q, r. So $p = 2$ or $p = 3$.

Case $p = 2$**.** Now $\frac{1}{q} + \frac{1}{r} = \frac{1}{2}$. Since $\frac{1}{r} \le \frac{1}{q}$,

$$\frac{1}{2} = \frac{1}{q} + \frac{1}{r} \le \frac{2}{q} \quad \Longrightarrow \quad q \le 4,$$

and since $\frac{1}{r} > 0$ we need $\frac{1}{q} < \frac{1}{2}$, so $q \ge 3$.

- $q = 3$: $\frac{1}{r} = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$, so $r = 6$.
- $q = 4$: $\frac{1}{r} = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$, so $r = 4$.

Case $p = 3$**.** Now $\frac{1}{q} + \frac{1}{r} = \frac{2}{3} \le \frac{2}{q}$, so $q \le 3$. With $q \ge p = 3$ this forces $q = 3$, and then $\frac{1}{r} = \frac{1}{3}$, so $r = 3$.

Every case has been covered, so the only solutions are

$$(p,q,r) = (2,3,6), \quad (2,4,4), \quad (3,3,3),$$

and there are no others.

Part (ii)

If $p = 1$ the sum is $1 + \frac{1}{q} + \frac{1}{r} > 1$ for every q and r. These are the solutions we're told about. (The ordering $p \le q \le r$ means these are $q = n$, $r = m$ with $1 \le n \le m$. The inequality the question writes between m and n is the wrong way round, but it doesn't matter: $p = 1$ works for any q and r.) So we look for solutions with $p \ge 2$.

As before, $\frac{1}{p}$ is the largest term, so

$$1 < \frac{1}{p} + \frac{1}{q} + \frac{1}{r} \le \frac{3}{p} \quad \Longrightarrow \quad p < 3.$$

With $p \ge 2$ this gives $p = 2$. We now need

$$\frac{1}{q} + \frac{1}{r} > \frac{1}{2}.$$

Ask yourself Why is the inequality now strict in the bound on q, and does it matter?

Since $\frac{1}{r} \le \frac{1}{q}$, we have $\frac{1}{2} < \frac{1}{q} + \frac{1}{r} \le \frac{2}{q}$, so $q < 4$. The strict inequality is what rules out $q = 4$. (In part (i), $q = 4$ was allowed, and gave the equality case $(2,4,4)$.) With $q \ge p = 2$ we get $q = 2$ or $q = 3$.

- $q = 2$: the sum is $\frac{1}{2} + \frac{1}{2} + \frac{1}{r} = 1 + \frac{1}{r} > 1$ for *every* r. So $(2,2,r)$ is a solution for every integer $r \ge 2$.
- $q = 3$: we need $\frac{1}{r} > \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$, so $r < 6$. With $r \ge q = 3$ this gives $r = 3,4,5$.

So, apart from the $p = 1$ family, the solutions are

$$(p,q,r) = (2,2,r) \text{ for any integer } r \ge 2, \qquad (2,3,3), \quad (2,3,4), \quad (2,3,5).$$

Each step above was forced, so there are no others.

Checks

Substitute directly. For example, $\frac{1}{2} + \frac{1}{3} + \frac{1}{5} = \frac{31}{30} > 1$, but $\frac{1}{2} + \frac{1}{3} + \frac{1}{6} = 1$ exactly. That shows why $(2,3,5)$ is the last of the $q = 3$ family, with $(2,3,6)$ the boundary case from part (i). Likewise $(2,4,4)$ gives exactly 1, so nothing with $p = 2$, $q = 4$ makes it into part (ii). A computer search over all $1 \le p \le q \le r < 400$ confirms both lists exactly.

Common slips

- Forgetting the infinite family $(2, 2, r)$ in part (ii). Once $p = q = 2$, the first two terms already add up to 1.
- Writing $p \leq 3$ instead of $p < 3$ in part (ii) and then wasting time on $p = 3$ (which gives a sum of at most 1).
- Stating the list of solutions without showing that every case has been checked. The "no others" is half the marks.

Beyond STEP

The solutions of $\frac{1}{p} + \frac{1}{q} + \frac{1}{r} > 1$ are exactly the triples for which a triangle with angles $\frac{\pi}{p}, \frac{\pi}{q}, \frac{\pi}{r}$ lives on a sphere. They match the finite symmetry groups in three dimensions: dihedral $(2, 2, r)$, tetrahedral $(2, 3, 3)$, octahedral $(2, 3, 4)$ and icosahedral $(2, 3, 5)$. Part (i) gives the three regular tilings of the flat plane. Equations like this, solved in integers, are Diophantine equations. Davenport's *The Higher Arithmetic* is a good introduction, and Allenby's *Rings, Fields and Groups* covers the symmetry groups.

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